

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Built production **LLM and agentic AI pipelines** (3 providers, 8+ models, Temporal.io durable workflows) at an AI startup; shipped observability tooling with OpenTelemetry traces and Prometheus metrics; benchmarked AI system behavior across hallucination rate, p50/p95 latency, and cost at **2,000 async req/sec**.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Built **multi-model LLM agent pipelines** across 3 providers and 8+ models (Anthropic Claude, OpenAI GPT-4o, Google Gemini) via MCP with **Temporal.io durable workflow orchestration** - provider-agnostic interfaces, structured outputs, agentic tool use.
- Shipped 10+ features end-to-end: TypeScript, Node.js, Express, Prisma ORM, **PostgreSQL**, **Redis**, Auth0, Docker, Sentry; real-time event pipelines and REST APIs serving production users across 7 deployments.

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- AI-driven claim automation (OCR, NLP, ML) improving accuracy 25%; automated AWS and GCP deployments with Grafana + Sentry observability.

PROJECTS

LLM Eval Harness 2025

Vendor-neutral benchmark evaluating AI system behavior: Claude, GPT-4o, and Gemini on 2,000-example frozen test set. Tracked **grounded hallucination rate**, **p50/p95 latency**, **cost/call**, and abstention accuracy with bootstrap significance testing + McNemar test.
Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

AlgoPulse -- Agentic AI Platform 2025

Event-driven LLM orchestration platform. Diagnosed API-polling bottleneck; pivoted to Supabase Edge Functions streaming AI reasoning in real time. **Temporal.io** durable workflow state + **Kafka** fault-tolerant coordination -- **2,000 async req/sec** without state loss.
TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

PulseRank -- Real-Time Leaderboard & Activity Feed 2026

Event-driven platform with full **observability stack**: **OpenTelemetry traces** + **Prometheus metrics** on all hot paths. MySQL outbox pattern, MongoDB TTL projections, Redis sorted sets for sub-ms reads, Server-Sent Events for live updates. Benchmarked with k6; GitHub Actions CI/CD.
TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, OpenTelemetry, Prometheus, Docker

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

AI, Operating Systems, Algorithms, Data Science, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | Machine Learning, DS & Algorithms, Big Data Infra, Data Warehousing

SKILLS

- Languages:** TypeScript, Python, JavaScript, Java, SQL, C
- AI/ML:** Anthropic Claude API, OpenAI API, Gemini API, LangChain, Hugging Face, PyTorch, RAG, MCP, LLM Evals
- Stack:** Node.js, React, NestJS, Next.js, Express, Prisma ORM, Temporal.io, Kafka, FastAPI
- Infra:** AWS, GCP, Docker, PostgreSQL, Redis, MongoDB, OpenTelemetry, Prometheus, Sentry, GitHub Actions
- Awards:** HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published, Top 1% Sanjeevani (1100+ teams)